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ARTICLE

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The effects of alternating tangential flow (ATF) residence time, hydrodynamic stress, and filtration flux on high-density perfusion cell culture

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Abstract

In this study, we investigated the effects of alternating tangential flow (ATF) cell separation on high-density perfusion cultures. We have developed methods to estimate theoretical residence times of cells in the ATF system and discovered that long residence times (above 75 s) correlate with decreased growth, metabolism, and productivity. We have calculated energy dissipation rates in the ATF transfer line and filter and empirically studied the impacts of increased exchange rates on cell culture, determining that increased hydrodynamic stress can lead to decreased cell size, lactate production, and specific productivity. Finally, we have conducted experiments to understand the relationship between filtration fluxes and ATF membrane fouling, finding that at fluxes above 60 $L{\cdot}m^{-2}{\cdot}day^{-1}$, protein sieving coefficients see significant rates of decrease (greater than 1% per day). While most of these studies have been conducted with one cell line at one target viable cell density (40 million cells/ml), the general, directional knowledge arising from this study should be applicable to other conditions and programs, ultimately leading to more robust and well-designed perfusion processes.

KEYWORDS
alternating tangential flow, ATF, energy dissipation rate, filtration flux, high cell density, perfusion, residence time, shear

1 | INTRODUCTION

Perfusion cell culture is a long-studied technique in the biotherapeutics industry (Castilho, 2014; Chang, Yoo, & Kim, 1994; Chotteau, 2015; Patil & Walther, 2016; Voisard, Meuwly, Ruffieux, Baer, & Kadouri, 2003; Woodside, Bowen, & Piret, 1998). Advantages relative to fed-batch culture include longer durations and fewer turnaround operations, higher cell densities and productivities, a healthier and more uniform culture across time, and potentially higher and more consistent product quality (Konstantinov & Cooney, 2015; Walther et al., 2018). Disadvantages include larger media volumes, challenges around long-term operability and sterility, longer development timelines, and a large amount of preexisting fed-batch infrastructure across the industry (Croughan, Konstantinov, & Cooney, 2015).

As the cell densities and productivities of perfusion systems increase, the case for continuous cell culture becomes more compelling (Pollock, Ho, & Farid, 2013). Economic analysis has predicted that at 120 million cells/ml there is a strong case for perfusion relative to fed-batch (Walther et al., 2015). Perfusion at the N-1 step can enable high-density inoculations, removing the need for long growth times to reach intensified target densities in the production bioreactor (Pohlscheidt et al., 2013; Wright et al., 2015; Yang et al., 2014). The cell density ceiling of a culture is, of course, linked to the particular cell separation technology used to enable perfusion. Most approaches relying on density-based separation (gravitational, centrifugal, and acoustic settling, for example) have been typically demonstrated only to 20–30 million cells/ml (Dalm, Cuijten, van Grunsven, Tramper, & Martens, 2004;

Kim et al., 2015). Early filtration approaches, such as spin filtration, are typically limited to peak densities around 30 million cells/ml, and filter fouling is a limiting factor (Kim et al., 2015; Patil & Walther, 2016).

In the past 5–10 years, however, tangential flow filtration and alternating tangential flow filtration (ATF) approaches have emerged as strategies that could break these previous cell density thresholds. These methods rely on a common principle to enable filtration even at high densities: Large tangential flows through a hollow-fiber filter prevent cells and debris from accumulating and fouling the filter, while a much smaller cross flow is pulled through the fiber and sent downstream as harvest. Peak densities of over 100 million cells/ml have been demonstrated (Clincke et al., 2013; Kim et al., 2015), and cultures above 40 million cells/ml have been sustained for long durations (Karst, Serra, Villiger, Soos, & Morbidelli, 2016; Martin et al., 2015; Warikoo et al., 2012; Xu & Chen, 2016). If additional intensification continues to occur and these high-density approaches prove translatable to commercial manufacturing, perfusion could become economically justifiable even for stable molecules traditionally manufactured with fed-batch bioreactors.

We have used the ATF in our efforts to intensify the production bioreactor (Godawat, Konstantinov, Rohani, & Warikoo, 2015; Warikoo et al., 2012; Wright et al., 2015). But as cell densities continue to be pushed higher, the cell separation device eventually will become limiting. An understanding of the ATF and the different stresses it exerts on cells and products is essential to a well-designed and robust process. Because the ATF and bioreactor are operated as an integrated unit, it is important to identify the critical ATF parameters and their interactions with bioreactor performance so that an acceptable overall design space can be defined for the combined unit operations.

In this manuscript, we build off of previous studies into ATF-related shear (Karst et al., 2016) and filter fouling (Bolton & Apostolidis, 2017; Hadpe, Sharma, Mohite, & Rathore, 2017; Kelly et al., 2014; Wang et al., 2017) and consider three potential effects of the ATF on the cell culture process: (a) cellular residence time in the ATF device, (b) hydrodynamic stress on cells within the transfer line and hollow fibers, and (c) concentration polarization and fouling of the hollow-fiber filter to proteins (and more specifically, the product). We use a combination of empirical and theoretical approaches to develop understanding around these effects and define safe operating spaces for the process.

2 | MATERIALS AND METHODS

A schematic of the ATF (Repligen, Waltham, MA) is shown in Figure 1. A diaphragm pump at the base of the device alternatingly draws in cell culture fluid and then returns it to the bioreactor. The fluid is pulled first through a transfer line and next through the lumina of a hollow-fiber filter and into the bulb of the ATF, after which it is returned through the same route back to the reactor. Cell-free perfusate is drawn perpendicularly through the hollow fibers using an external pump. The exchange rate of this alternating flow is high relative to the perfusion rate, creating a significant backflush and allowing the system to perform for longer durations without filter fouling. In this manuscript, we use "downward half-cycle" to refer to the transition from a fully inflated to fully deflated diaphragm and "upward half-cycle" to refer to the opposite transition (fully deflated to fully inflated). One cycle comprises a downward half-cycle followed by an upward half-cycle.

We coupled stirred-tank bioreactors (Broadley-James, Irvine, CA) with ATF filtration devices to culture Chinese hamster ovary (CHO) cells engineered to secrete recombinant proteins. The standard setup was a bioreactor connected to an ATF filtration device to enable perfusion (Figure S1 in Supporting Information Materials). Fresh medium was fed into the reactor at a constant rate after inoculation. Cell-free harvest was removed through the ATF at a rate necessary to keep the bioreactor weight (and therefore volume) constant. Once the bioreactor achieved its target production cell density, cell-containing fluid was removed directly from the bioreactor at a rate such that the cell density was kept constant. Aber Futura probes (Aber Instruments, Aberystwyth, UK) measured capacitance as an online proxy for cell density. In some instances, if capacitance probes were not available, cells were bled in daily boluses to keep the culture near the target density. Cell densities were typically controlled around 40 million cells/ml.

Perfusion was conducted with chemically defined media (1–2 reactor volumes/day) with one cell line producing a recombinant enzyme (cell line A) and two cell lines producing monoclonal antibodies (cell lines B and C). Glass bioreactors with 3.75 and 10 L working volumes and single-use bioreactors with 50 L working volumes were paired with ATF2, ATF4, and ATF6 devices, respectively. Polyethersulfone filters (Repligen) composed of fibers with 1 mm lumen diameters and 0.2 μm pores were used in the ATFs. Sparged oxygen (sintered or fritted microspargers) was used to control dissolved oxygen levels. Viable cell density and viability were measured with Vi-CELL (Beckman Coulter, Brea, CA), and cell culture indicators such as glucose, lactate, and lactate dehydrogenase (LDH) activity were measured with the Cedex Bio HT (Roche Diagnostics, Indianapolis, IN). Antibody product concentrations were also measured with the Cedex Bio HT, while enzyme product activities were measured using an internal activity assay.

2.1 | ATF residence time

Cell residence time in the ATF should be minimized to reduce exposure to the uncontrolled environment of the ATF. This residence time is primarily impacted by the ATF exchange rate and the ATF dead volume (combined volume of the transfer line and filter). The actual exchange rate can differ from the inputed exchange rate set point, because the timing of the diaphragm pump is based on an assumed pump volume, when in reality, the pump volume actually increases

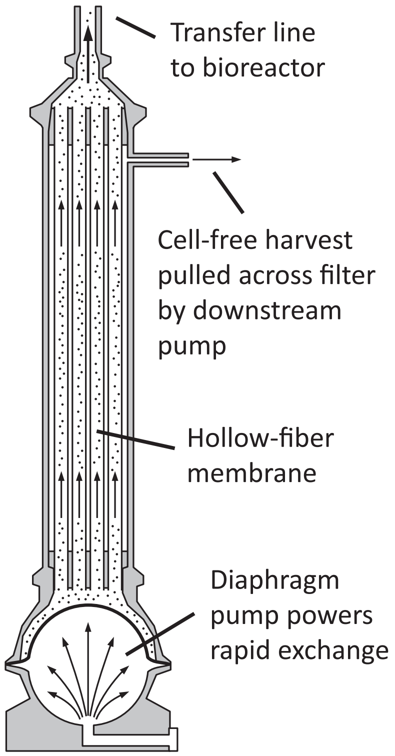

FIGURE 1 Diagram of the ATF showing that exchange flow is powered by a diaphragm pump, alternatingly sending cell culture fluid to and from the bioreactor, up and down through the lumina of a hollow-fiber membrane. Meanwhile, an external pump pulls cell-free harvest perpendicularly across the filter

with exchange rate (since the silicone diaphragm stretches more at higher rates). We account for this with a simple proportional correction:

$$Q = Q_{\mathrm{input}} \cdot \frac{V_{\mathrm{pump}}}{V_{\mathrm{pump,0}}}, \tag{1}$$

where Q_{input} is the inputed exchange rate, Q is the actual exchange rate, $V_{\mathrm{pump,0}}$ is the assumed pump volume, and V_{pump} is the actual, empirically measured pump volume corresponding to Q_{input}.

The primary concern around long residence times in the ATF is oxygen availability. We can approximate the time until dissolved oxygen is depleted in the ATF by dividing the oxygen concentration by the oxygen consumption rate:

$$t_{\mathrm{deplete}} = \frac{c_{O_2} \cdot f_{O_2}}{q_{O_2} \cdot x_v}, \tag{2}$$

where t_{deplete} is the time until oxygen depletion, c_{O_2} is the molar solubility of oxygen in the cell culture fluid in an air environment, f_{O_2} is the level of controlled dissolved oxygen in the reactor (where a value of one corresponds to maximum solubility at a dissolved oxygen set point of 100%), q_{O_2} is the cell-specific molar oxygen uptake rate, and x_v is the viable cell density. The molar solubility of oxygen can be estimated using a combination of the Setschenow and van't Hoff relationships (Weiss, 1970).

We next must estimate the residence time of cells in the ATF and compare that to the oxygen depletion time. To arrive at an approximation, we make several initial assumptions: (a) fluid in the transfer line and hollow fibers travel at average velocities that do not vary with radial position; (b) negligible axial mixing occurs in the transfer line and hollow fibers; (c) perfect mixing occurs in the ATF bulb at the conclusion of every downward half-cycle; and (d) all cells transferred into the ATF during one cycle spend the entire duration of the downward half-cycle in the ATF (Figure 2a).

With these assumptions in place, we observe that cells that end in the transfer line or filter at the end of the downward half-cycle will be returned immediately to the bioreactor at the beginning of the upward half-cycle, and therefore will experience negligible residence times in the ATF. However, cells that reach the bulb during a downward half-cycle can potentially remain within the ATF for several cycles, especially as the ratio of the dead volume to pump volume increases. To estimate the residence time in the ATF system, we begin by calculating the fraction of the bulb volume that is exchanged with fresh bioreactor culture after each cycle as follows:

$$f_{\mathrm{exch}} = \frac{V_{\mathrm{pump}} - V_{\mathrm{xfer}} - V_{\mathrm{filter}}}{V_{\mathrm{bulb}}}, \tag{3}$$

where f_{exch} is the fraction of the bulb volume exchanged with fresh bioreactor culture (i.e., the exchange fraction) and V_{pump}, V_{xfer}, V_{filter}, and V_{bulb} are the ATF pump, transfer, filter and bulb volumes, respectively (see Figure S2 in Supporting Information Materials for a schematic describing these volume definitions). Equation (3) shows that fluid can only return to the bioreactor (and be exchanged) inasmuch as the pumped volume overcomes the dead volume in the filter and transfer line. We can next calculate the probability that a cell is trapped in the ATF bulb for a given number of cycles using the expression

$$p = f_{\mathrm{exch}} \cdot (1 - f_{\mathrm{exch}})^{n-1}, \tag{4}$$

where p is the probability that a cell is in the ATF for n cycles. An average number of cycles experienced by a cell in the ATF can then be calculated using the probability distribution in Equation (4). This weighted average is a power series that converges simply to the inverse of f_{exch}:

$$\sum_{n=1}^{\infty} n \cdot f_{\mathrm{exch}} \cdot (1) - f_{\mathrm{exch}})^{n-1} = f_{\mathrm{exch}}^{-1}, \tag{5}$$

where $0 \le f_{\mathrm{exch}} \le 1$. The average residence time can then be determined with the following simple expression:

$$\tau = \frac{t_{\mathrm{cycle}}}{f_{\mathrm{exch}}}, \tag{6}$$

where τ is the average residence time in the ATF system and t_{cycle} is the ATF cycle time:

$$t_{\mathrm{cycle}} = \frac{V_{\mathrm{pump}}}{Q}. \tag{7}$$

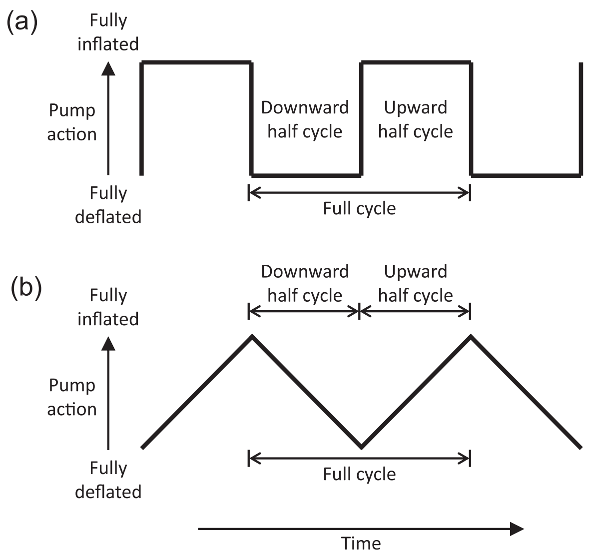

FIGURE 2 ATF cycle assumptions for (a) the simplified approach outlined in Equations (3–6) and (b) the Monte Carlo approach

Note that most previous, conventional ATF residence time calculations assume that the entire ATF system is well mixed. Under this assumption, the fraction of the system that is returned to the bioreactor after each cycle can be calculated with the following expression:

$$f'_{\text{exch}} = \frac{V_{\text{pump}}}{V_{\text{bulb}} + V_{\text{xfer}} + V_{\text{filter}}}. \tag{8}$$

Residence time distributions and average residence times can then be calculated as before using Equations (4)–(6). However, this approach leads to dramatically reduced residence time calculations and can underestimate the impact of dead volume, which is why we advocate the use of the assumption of only a well-mixed bulb and the resulting exchange fraction expression in Equation (3).

Some of the assumptions building up to the residence time approximation in Equation (6) are inaccurate and can be improved. In particular, the hollow-fiber diameter in the filter is small enough such that flow can be assumed to be laminar, leading to negligible radial mixing and an axial velocity that depends on radial position. Also, the ATF flow does not occur instantaneously at the beginning and end of each cycle, but instead is fairly evenly distributed across the entire cycle (depicted in Figure 2b). We can incorporate these additional refinements into an improved residence time calculation by simulating random particles that reach the ATF and combining these simulations into a Monte Carlo approach to get residence time distributions and average residence times (see Figure S3 in Supporting Information Materials for a flow chart and equations illustrating the simulation approach). Outputs are average residence time and residence time distribution. The simulations were written and executed using the Python programming language.

We applied both the simplified model (Equations (3)–(7)) and the Monte Carlo model (Figure S3 in Supporting Information Materials) to estimate ATF residence times for a number of previous actual perfusion bioreactor runs that had been conducted across a variety of cell lines and scales. These residence times were then correlated with run success, which is defined qualitatively as the system's ability to maintain a positive apparent growth rate while the culture is maintained at a target cell density (via cell bleed) for a prolonged duration.

2.2 | Hydrodynamic stress in the ATF

Energy dissipation rate (EDR) has also been discussed as a good way of predicting cell damage in a system (Godoy-Silva, Mollet, & Chalmers, 2009; Hu, Berdugo, & Chalmers, 2011; Karst et al., 2016; Mollet et al., 2004). Because of the high exchange rate through the ATF, it is possible that cells traveling through the transfer line and the hollow fibers could experience high levels of shear that could negatively impact the culture.

We can calculate EDRs in the transfer line and hollow-fiber lumina first by calculating the Reynolds number (Re) in these different regions of the device. For regions where Re is less than 2,100, we can assume laminar Hagen–Poiseuille flow, in which case EDR varies across the radius according to the following expression (Mollet et al., 2004):

$$\epsilon = \frac{16\mu v^2 r^2}{R^4}, \tag{9}$$

where ϵ is the EDR, μ is the viscosity of the fluid, v is the average fluid velocity, R is the radius of the tube, and r is the radial position within the tube. The maximum EDR occurs at the tube wall and the average EDR is half of the maximum. Note that a substitution has been made to Mollet's original equation, so that Equation (9) is in terms of velocity rather than volumetric flow.

When Re is greater than 4,000, turbulent flow can be assumed. To obtain EDR in a tube with a smooth surface (e.g., silicone) in this flow regime, the Fanning friction factor first must be calculated (Bird, Stewart, & Lightfoot, 2002):

$$f = \frac{0.0791}{Re^{0.25}}. \tag{10}$$

The volumetric EDR in a tubular system (due to friction) is a product of the pressure drop and the volumetric flow (Cengel & Cimbala, 2006) divided by the volume under consideration:

$$\epsilon = \frac{Q\Delta P_L}{V} = \frac{Av\Delta P_L}{AL} = \frac{v\Delta P_L}{L} \tag{11}$$

where Q is the volumetric flow through the tube, V is the volume of the tube over a length L, A is cross-sectional area of the tube, and ΔP_L is the pressure drop over a length L of the tube. The pressure drop can be calculated using the friction factor:

$$\Delta P_L = \frac{fL\rho v^2}{R}. \tag{12}$$

By combining Equations (11) and (12), we can arrive at an expression for EDR in turbulent flow:

$$\epsilon = \frac{f\rho v^3}{R} \tag{13}$$

For the purposes of the simulations in this study, we assume that turbulent flow occurs in the transfer line and laminar flow in the hollow fibers. This generally holds true, with the caveat that for some combinations of low exchange rates and large-diameter tubing, laminar flow does exist in the transfer line. Equations (9) and (13) therefore allow us to calculate the "instantaneous" EDR experienced by cells in the filter and transfer line, respectively. We also assume that the EDR in the ATF bulb is negligible. To gain a more holistic understanding of EDR experienced by cells over each visit to the ATF, we can couple the Monte Carlo approach mentioned in the previous section with EDR calculation. For each Monte Carlo simulation, we track the location, duration, and EDR experienced by a particle after it enters the ATF. With this framework in place, we can generate cumulative distributions of EDRs experienced by cells in the ATF system.

To combine this theoretical approach with experimental data, univariate ATF4 studies were also conducted in benchtop bioreactors where exchange rate was varied within a run from 2 to 8 L/min and cell responses (growth, metabolism, and productivity) were monitored. At the end of the run, exchange rate was returned to its original value to confirm that hysteresis effects were minimal. These studies give information that can explain how cell culture responds to increased shear in the ATF.

2.3 | ATF filter fouling

Two replicate bioreactor/ATF systems were operated at various filtration fluxes and protein sieving was measured across the filter to gauge the effect on filter fouling. To understand the impact of ATF scale on filter fouling, each bioreactor was linked with an ATF2 and ATF4 (both simultaneously running). Cell density was controlled at approximately 40 million cells/ml. The filtration rates exiting each ATF were chosen so that filter fluxes (volumetric flow per filter surface area) were equal across ATF scales. To ensure that cells in the bioreactor experienced the same environment across the experiment, the overall perfusion rate was maintained at 2 RV/day. At higher filtration fluxes, this was accomplished with a cell-free recirculation loop to send a portion of the post-ATF4 harvest material back to the reactor (see Figure S4 in Supporting Information Materials for a schematic of the operation). Exchange rates were held constant at 1 and 3.5 L/min for the ATF2 and ATF4, respectively. Across one perfusion run, five different filtration fluxes were successively studied. ATFs were replaced after the third flux condition to avoid excessive fouling. Each flux condition was held for 5–20 days during which sieving coefficients for two proteins (the product and lactate dehydrogenase) were calculated using the following expression:

$$\sigma = \frac{c_h}{c_r}, \tag{14}$$

where σ is the sieving coefficient, c_h is the protein concentration (or activity) in the harvest (post-ATF), and c_r is the protein concentration (or activity) in the bioreactor (pre-ATF). Sieving coefficients were considered to be primary indicators of ATF filter fouling: A sieving coefficient near one indicates a clean, nonfouled filter while a sieving coefficient of zero indicates that no product molecules can traverse the filter.

To compare different bioreactor/ATF conditions, we measured sieving coefficients over time for each condition and calculated and compared the slopes of best-fit lines for each of those profiles. These slopes indicate the rate of filter fouling for the different conditions.

To verify that filter history did not confound the filter fouling rate measurement, we calculated a theoretical exposure value for the filter over time for use as another possible factor in statistical analysis:

$$E = \int j \cdot c_r \cdot \mathrm{d}t, \tag{15}$$

where E is exposure, j is the volumetric filter flux at time t, and c_r is the concentration of a protein in the bioreactor that is indicative of the total load on the filter from the bioreactor. We used both LDH and the product to calculate exposures, and calculated the integral in Equation (15) using the trapezoid method for the discrete time points.

A data table containing categorical factors (ATF type, bioreactor replicate), continuous factors (filtration flux, exposure), and responses (LDH and product sieving coefficient rates of change) was compiled and analyzed in JMP (SAS Institute, Cary, NC) to understand which factors were statistically and practically significant. We used stepwise regression with a p-value threshold stopping rule (0.1 to enter and 0.05 to leave).

3 | RESULTS AND DISCUSSION

3.1 | ATF residence time

ATF-related problems were originally observed when comparing cell culture performance in 4 L/ATF2 and 10 L/ATF4 systems. The ATF4 systems were successfully operated for 60 days at 40 million cells/ml while the ATF2 systems could not maintain cell growth at 30 million cells/ml, leading to decreased cell density and early conclusion of the experiment (Figures 3a,b). Cell-specific productivity also dropped precipitously and cell-specific lactate increased (Figures 3c,d). Examination of the 2 L/ATF2 systems and residence time calculations showed that large ATF2 transfer line volumes led to low exchange fractions and much longer residence times (90–190 s in the ATF2 vs. 70 s in the ATF4).

Oxygen depletion time was calculated to be 38 s for these conditions (using Equation (2)), significantly lower than the ATF2 residence times, suggesting that prolonged exposure of the culture to the uncontrolled environment of the ATF bulb could be pushing cells into hypoxic conditions ultimately leading to a culture crash. To test this hypothesis, the dissolved oxygen in the bioreactor was increased from 40% to 70% in one midcrash 2 L/ATF2 system. Growth, viability, metabolism, and productivity all stabilized (Figure 4a,b,c,d), indicating that the negative effects of longer residence times could be countered by increasing the amount of available dissolved oxygen in the system.

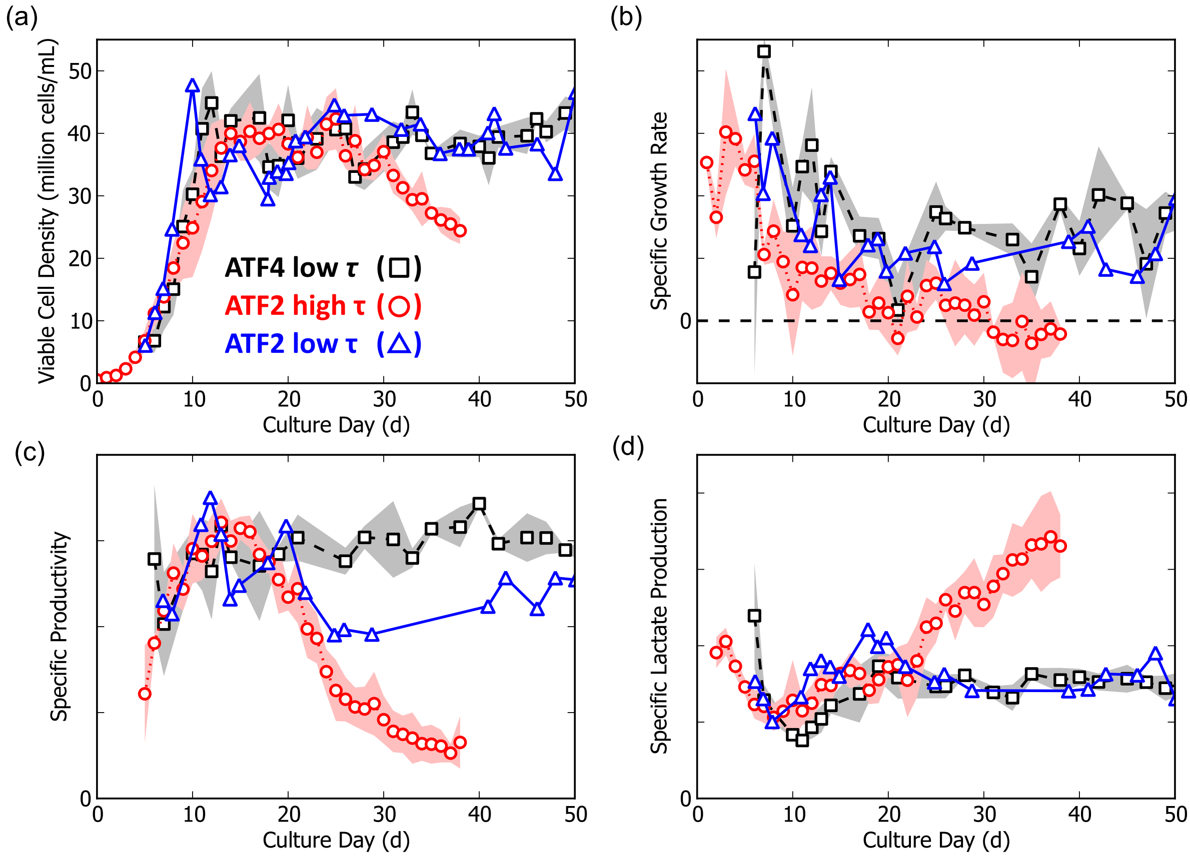

FIGURE 3 (a) Viable cell density, (b) specific apparent growth rate, (c) specific productivity for the protein of interest, and (d) specific lactate production for ATF4 ($n = 3$) and ATF2 systems ($n = 1$) with low residence times and an ATF2 system ($n = 4$) with high residence times. All runs were conducted at 40% dissolved oxygen. Shaded regions represent ±1 *SD*. [Color figure can be viewed at wileyonlinelibrary.com]

Note that this study of the impact of increased dissolved oxygen was actually conducted as an end-of-run study. The bioreactor had previously grown up to a target density and was then bled to approximately 4 million cells/ml on Day 45 to attempt a new experiment. To enable comparison with the aforementioned ATF2 experience, the plotted run day for this study is shifted by 40 days.

To bring the ATF2 residence times in line with the ATF4 residence time, the ATF2 transfer line was redesigned with narrower (6.4 vs. 9.6 mm inner diameter) tubing. Successful runs were then demonstrated at the standard 40% dissolved oxygen set point. One of these successful ATF2 runs is overlaid in Figure 3. Cell density and cell growth remain robust through 50 days, and specific lactate production also remains low. Specific productivity does not crash like previous ATF2 runs. An offset in specific productivity between the 10 L/ATF4 system and the improved ATF2 system still does remain, as seen in Figure 3c. We hypothesize this is due to other to-be-solved scale-up considerations, and not to ATF residence time.

To further explore the relationship between ATF residence time and run success, we have conducted a data mining exercise and pulled together experiences across different cell lines and different bioreactor and ATF scales. Using dead volume measurements and exchange rate data, we have calculated theoretical ATF residence times using both the simplified model as well as the Monte Carlo approach. In either case, we observe a strong correlation between longer residence times and run failure (Table 1).

The Monte Carlo simulation data for a particular ATF can be presented as a residence time distribution. As an example, the residence time distribution for an ATF4 system with an exchange rate at 3.5 L/min is shown in Figure 5a. The majority of the cells that enter the ATF system only reach the transfer line or filter, and return to the bioreactor within a single cycle. However, there is a long tail of cells that reach the bulb and are trapped there for one or more cycles. The average residence times in Table 1 are calculated from this CRBALO (cells reaching bulb at least once) population.

We can also compare residence time distributions between different ATF scenarios. In Figure 5b, we overlay the residence time distribution resulting from typical ATF4 operation to a distribution from an ATF2 with long residence times. Only CRBALO residence times are considered. We can see that the long-residence-time scenario is impacted by both the low exchange rate (leading to fewer opportunities for cells to escape the bulb) as well as the larger dead volume (leading to a smaller fraction of cells in the bulb to leave with each cycle).

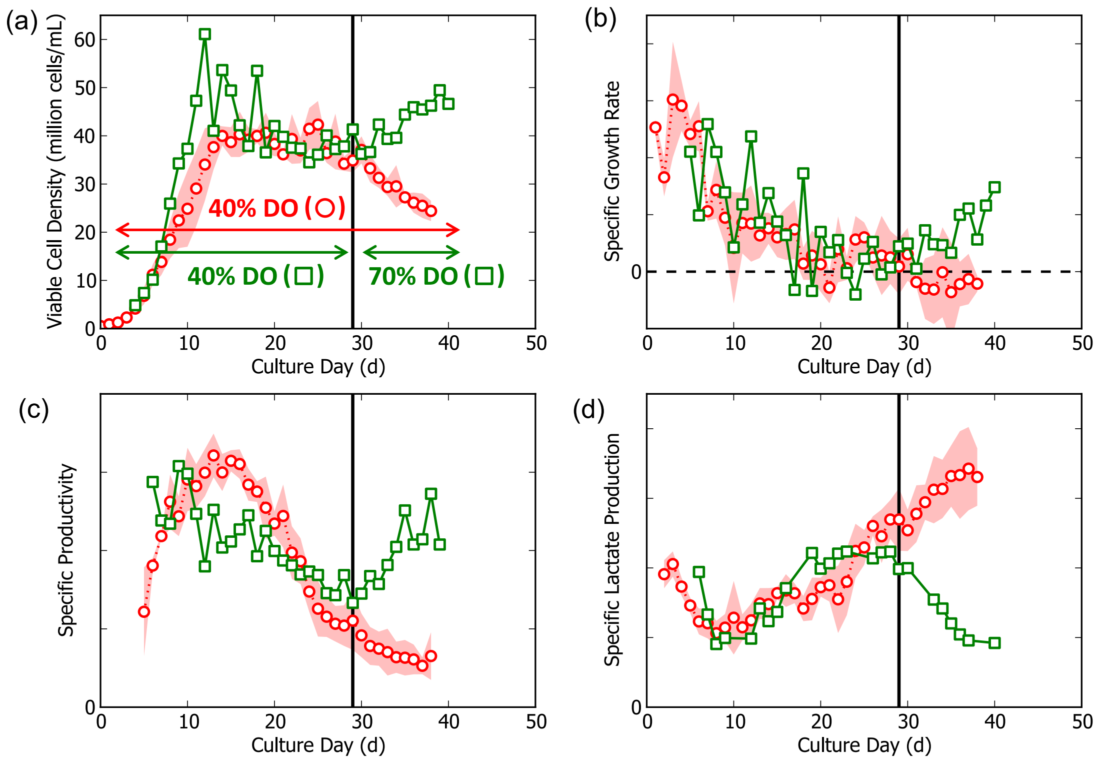

FIGURE 4 (a) Viable cell density, (b) specific apparent growth rate, (c) specific productivity for the protein of interest, and (d) specific lactate production for an ATF2 system with a midrun DO shift (40–70%), reversing negative growth and metabolic trends. The baseline ATF2 profile (40% DO) is shown for comparison. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 ATF runs and calculated residence times

Working volume (L)	Cell line	ATF type	Exchange rate (L/min)	Cycle time (s)	Exchange fraction	τ simplifed (s)	τ M. Carlo (s)	Run success
2	A	ATF2	0.5	23.8	0.13	176.6	205.8	No ✘
3.75	A	ATF2	0.4	29.7	0.20	150.7	185.8	No ✘
10	A	ATF4	2	25.2	0.24	106.0	121.8	No ✘
2	A	ATF2	0.8	14.9	0.15	99.8	121.2	No ✘
4	A	ATF4	2.3	21.9	0.25	88.8	105.3	No ✘
2	A	ATF2	1.1	10.8	0.18	59.1	77.9	No ✘
40	A	ATF6	10	14.4	0.20	71.5	77.7	Yes ✓
50	A	ATF6	11	13.1	0.21	61.6	68.9	Yes ✓
4	A	ATF4	3.5	14.4	0.32	45.4	60.3	Yes ✓
10	A	ATF4	3.5	14.4	0.33	44.2	59.1	Yes ✓
3.75	B	ATF2	0.8	14.9	0.33	45.4	57.8	Yes ✓
3.75	A	ATF2	1	11.9	0.34	35.0	45.6	Yes ✓
500	C	ATF10	60	12.0	0.33	35.9	39.9	Yes ✓
10	A	ATF4	5	10.1	0.37	27.2	38.5	Yes ✓
3.75	B	ATF2	1.2	9.9	0.38	25.9	34.7	Yes ✓
10	A	ATF4	6.5	7.8	0.41	18.7	27.7	Yes ✓
10	A	ATF4	8	6.3	0.46	13.8	21.3	Yes ✓

Note. Residence times are color coded, where red and green indicate longer and shorter durations, respectively. [Color figure can be viewed at wileyonlinelibrary.com]

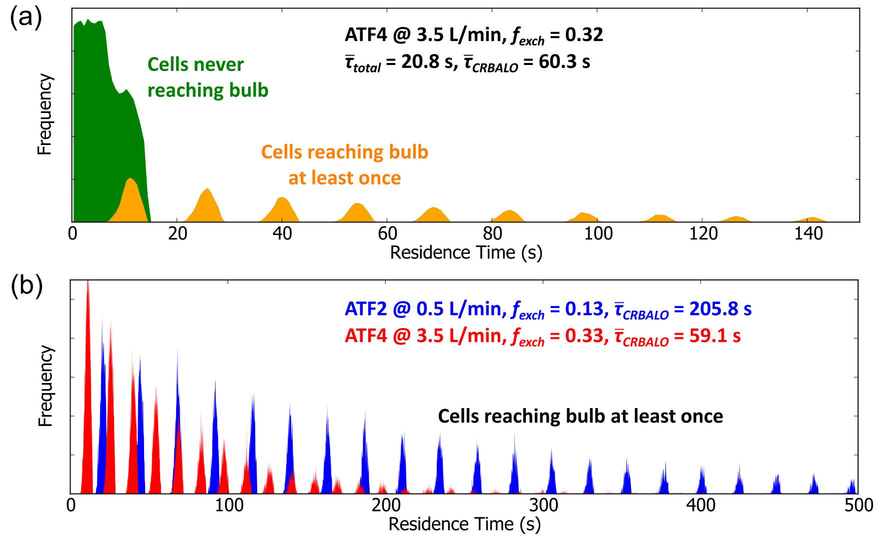

FIGURE 5 (a) Residence time distribution for an ATF4 with an exchange rate of 3.5 L/min and an exchange fraction of 0.32. The histogram is divided into areas representing cells that never reach the bulb and cells that reach the bulb at least once (CRBALO). (b) CRBALO residence time distributions for two scenarios: An ATF2 with a large dead volume and slow exchange rate and an ATF4 with a small dead volume and moderate exchange rate [Color figure can be viewed at wileyonlinelibrary.com]

3.2 | Hydrodynamic damage in the ATF

We used typical ATF line specifications (Table SI in Supporting Information Materials) along with Equations (9) and (13) to calculate maximum EDRs in the transfer line and hollow-fiber filter for the ATF2, ATF4, ATF6, and ATF10. EDRs across the exchange rate operating ranges for each of these devices are plotted in Figure 6a. In the ATF2 and ATF6, the greatest stress typically falls within the filter. For the ATF4, however, because the filter has approximately twice the number of fibers per volumetric flow, the hollow-fiber EDR is significantly lower than in the other ATFs. The ATF10 also sees high EDR in its transfer line, because here we assume a two-line configuration, increasing the wall area.

It is important to note that all of the EDRs across all ATFs and exchange rates fall well below typical values for lethal EDRs (10^6 W/m^3 and greater) reported in the literature. However, they do fall within reported ranges for sublethal responses (Chalmers & Ma, 2015). In one previously reported experiment, CHO cells continuously exposed to EDRs of only 925 W/m^3 (corresponding to a shear stress of 0.8 Pa) in a laminar flow chamber demonstrated decreased productivity and lactate yield from glucose (Keane, Ryan, & Gray, 2003). It should be noted that (a) these cells were attachment-dependent and (b) they were continuously exposed to the stress unlike our perfusion system where cells visit the ATF for a few seconds every few minutes. Another study showed that EDRs as low as 400 W/m^3 could decrease specific productivity by 25% in a suspended CHO culture in bioreactors (Sieck et al., 2013). Other experiments showed changes in glycosylation patterns (but not growth and metabolism) at EDRs of 60,000 W/m^3 (Godoy-Silva et al., 2009). Clearly, hydrodynamic stress can significantly influence cells and product, but the threshold will vary based on cell line and the frequency and duration of exposure.

For our particular experimental system, we can combine EDR calculation with Monte Carlo simulation to generate theoretical EDR distributions. Cumulative distribution functions (CDFs) are one useful way to visualize the data and aid understanding. We can plot CDFs against time instead of probability to quickly show the durations that cells are spending at or above different EDRs. In Figure 6b, we plot time-based CDFs for the four ATF scales at typical exchange rate set points. We see that the ATF6 offers the least EDR exposure while the ATF2 and ATF10 have more stressful environments. Likewise, we can observe the increasing EDR profiles for the ATF4 in Figure 6c as the exchange rate is increased from 3.5 to 8 L/min. The two-phase patterns in Figures 6b,c are a result of the different EDR contributions from the hollow-fiber filters (lower EDRs, with a distribution of possible values depending on the particle's radial distance) and the transfer line (higher EDRs, with a single value for a given configuration due to the assumption of radial mixing).

For these types of plots to be useful, we need to understand when EDRs are biologically significant and can affect the cell culture or the therapeutic product. We conducted an experiment in a 10 L/ATF4 system where the exchange rate was varied stepwise across the course of a perfusion experiment, and cell growth, size, metabolism, and productivity were tracked. The cell culture was not viable at the lowest exchange rate (2 L/min), presumably because the long residence times in the ATF (calculated to be 140 s) led to hypoxia (Figure 7a). As the exchange rate increased above 5 L/min, lactate yield from glucose, cell-specific productivity, and viable cell diameter all decreased (Figure 7b,c, d). Specific growth rate did not noticeably change, but the noise in this measurement hampered our ability to draw firm conclusions. Referring back to Figure 6c, the maximum EDR experienced in that ATF4 at 5 L/min is approximately 15,000 W/m^3, indicating that this is the EDR threshold marking major shifts in metabolism and cell morphology for this particular cell line and system.

We should note, however, that this experiment does not allow us to decouple increasing EDR from increasing frequency of exposure to the ATF environment. As exchange rate increases from 3.5 to 8 L/min, the average residence time of a cell in the bioreactor (e.g., time between ATF visits) decreases from 2.9 to 1.3 min. In the absence of additional experimental data, we therefore need to take both EDR and frequency of exposure into consideration as we scale the system.

3.3 | ATF filter fouling

Protein sieving in ATF2s and ATF4s was tracked across five different filter fluxes ranging from 22 to 115 L·m^{-2}·day^{-1} in two replicate bioreactors. All of these experiments were conducted using one cell line at one target cell density. Since filter fouling rate can vary widely across different cell lines and cell densities, the results below cannot be considered universal, and new experimental data must be generated for different scenarios.

Sieving coefficient profiles and corresponding best-fit lines were built (using both product activity and LDH activity measurements) for each condition. Product and LDH profiles can be, respectively, found in Figures S5A and S5B for the ATF2 and Figures S5C and S5D for the ATF4 in the Supporting Information Materials. The slopes of these best-fit lines, which we use as a proxy for the rate of filter fouling, are summarized in Figure 8 and Table SII (in Supporting Information Materials). There is a large amount of noise in these calculations, especially for the slopes that needed to be fit over only short durations. For example, the 75 L·m^{-2}·day^{-1} condition actually shows increasing sieving coefficients for some of the ATFs, but this might be attributed to the short duration (7 days) for which this flux was held. Despite this potential noise, a general (and expected) trend does appear: As filter flux increases, filter fouling rate increases. In particular, Figure 8 demonstrates that at and above 90 L·m^{-2}·day^{-1}, filter fouling rates are practically significant (greater than 1% per day).

To statistically confirm the correlation with filter flux, and to understand if any other factors clearly contribute to filter fouling, we performed stepwise regression with JMP examining ATF type, bioreactor number, filter flux, and filter exposure. Quadratic and interactive effects for the two continuous factors (flux and exposure) were also examined. Out of all of these factors, only flux could be identified as significantly correlated to filter fouling, as measured by both product and LDH sieving. This statistical analysis results in a simple quadratic predictive expression:

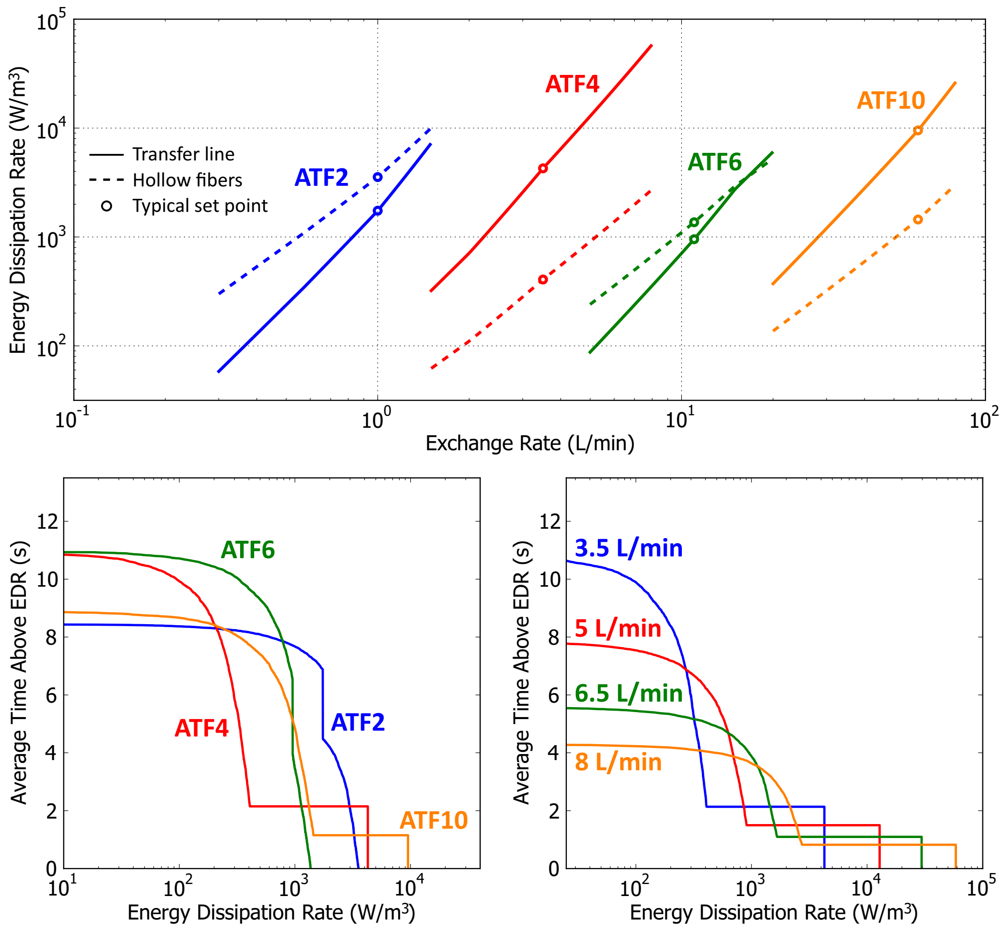

FIGURE 6 (a) Theoretical energy dissipation rates within the transfer line and hollow-fiber filter for various ATF sizes across their respective exchange rate ranges. Solid lines indicate the average EDR in the transfer line and dashed lines indicate the maximum EDR in the hollow-fiber filter. Cumulative EDR distributions are shown in (b) for the ATF2, ATF4, ATF6, and ATF10 at 1, 3.5, 11, and 60 L/min exchange rates, respectively, and (c) the ATF4 at exchange rates of 3.5, 5, 6.5, and 8 L/min. For the EDR distribution plots, the *y* value is the average time that a particle spends above or equal to the energy dissipation rate (indicated by the *x* value). Distributions with long and large tails will be potentially detrimental to cell health, metabolism, or productivity. Straight-line drops in the distributions occur at the transfer line EDRs, since all particles in well-mixed turbulent plug flow are assumed to see the same average EDR [Color figure can be viewed at wileyonlinelibrary.com]

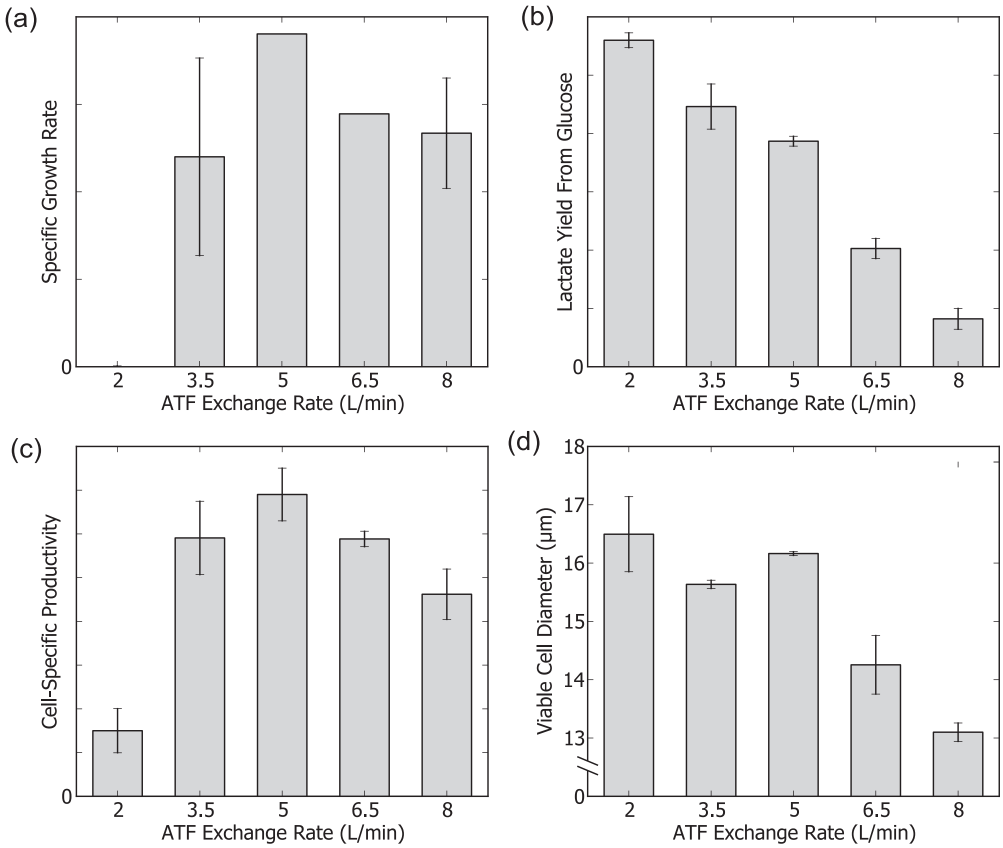

FIGURE 7 (a) Specific growth rate, (b) lactate yield from glucose, (c) cell-specific productivity, and (d) viable cell diameter in a 10 L/ATF4 system across various exchange rate set points

$$r_f = A{\cdot}j^2 + B{\cdot}j + C, \tag{16}$$

where r_f represents the filter fouling rate (as measured either by product or LDH sieving coefficient) and j is the filter flux. The coeffiecients for Equation (16) for the case of LDH and product are shown in Table SIII in Supporting Information Materials and the predicted profiles along with confidence intervals are shown in Figures S6A and S6B in the Supplementary Materials.

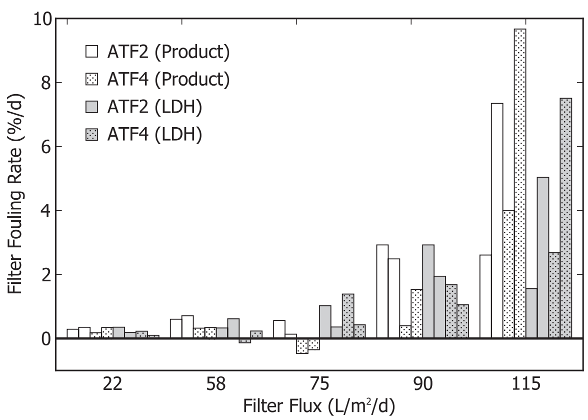

FIGURE 8 Estimated filter fouling rates across a range of filter fluxes using product activity and LDH activity measurements for ATF2 and ATF4 units. Replicate conditions were run, and both data sets are shown here separately

The magnitudes of the filter fouling are somewhat offset (when comparing LDH vs. product), but the confidence intervals between the two protein fouling rates overlap generously. The shapes of the profiles also agree closely, with little to no effect being seen on filter fouling up to 65–70 L·m^{-2}·day^{-1}. Above 70 L·m^{-2} ·day^{-1}, fouling rates increase significantly with the filter flux. If additional confidence is needed, more experiments would need to be executed using a design-of-experiments approach to improve resolution.

We can examine potential reactor/ATF/perfusion rate combinations to estimate when filter fouling might be of concern, at least for this particular cell line and target cell density. Table 2 lists filter fluxes for some typical situations. If we decide to continue to maintain filter fluxes under 60 L·m^{-2}·day^{-1}, we cannot run a 100 L bioreactor with only one ATF (80 L·m^{-2}·day^{-1}), but would instead need two simultaneously operating units (40 L·m^{-2}·day^{-1}). A similar approach would be necessary at the 500 L scale (two ATF10s instead of just one). The ATF approach in general also appears to be limited when considering 2000 L single-use bioreactors. Even a pair of ATF10s would lead to filter fluxes three times larger than the recommended maximum flux of 70 L·m^{-2}·day^{-1}.

Other studies show a wide variety of possible filter performances, with filter fouling rates ranging from less than 0.5%/day (Karst et al., 2016) to approximately 3%/day (Wang et al., 2017). As noted previously, it is of course important to remember that the data reported in this study is all generated for one cell line at one target cell density using one filter type. Different cell lines, cell densities, filter

TABLE 2 Hypothetical filter fluxes for various combinations of ATF, bioreactor, and perfusion rate

ATF type	Bioreactor volume (L)	Filter flux ($L/m^2/d$) @ 1 RV/d	Filter flux ($L/m^2/d$) @ 2 RV/d
ATF2	2	15	29
ATF2	4	29	58
ATF4	10	13	26
ATF6	50	20	40
ATF6	100	40	80
ATF10	500	48	95
2XATF10	2000	95	190

Note. Filter fluxes are color coded, where red and green indicate higher and lower fluxes, respectively. [Color figure can be viewed at wileyonlinelibrary.com]

types, and operational set points (e.g., exchange rates) could lead to different fouling trends. Also, product stability (or lack thereof) could influence acceptable sieving coefficient ranges. Products with relatively good stability will be able to better withstand low sieving coefficients (and therefore longer durations in the bioreactor before passing through the filter as downstream harvest).

4 | CONCLUSION

The ATF considerations that have been discussed in this manuscript involve many different operating parameters, leading to a complex, hyperdimensional design space with various constraints. For example, filter fouling is a function of multiple parameters, including exchange rate, perfusion rate, filter area, and cell density. The ATF residence time (and its acceptability) is also a function of exchange rate, but in addition, depends on the cell density, the dissolved oxygen level, and the ATF dead volume. Hydrodynamic stress depends on exchange rate and the dimensions of the transfer line and filter, while its severity will be a function of the cell line. Figure S7 (in the Supporting Information Materials) qualitatively summarizes some of these ideas, indicating that these constraints must be navigated to find acceptable operating set points and ranges.

We can synthesize the results and conclusions from the previous sections to form a general approach for ATF scale-up or scale-down:

1. Find a combination of bioreactor working volume and ATF that lead to an approximate match for filter flux. In the case described in the filter fouling section, the upper limit would be approximately 60 $L\cdot m^{-2}\cdot day^{-1}$, but for other situations, it will be variable and will depend on several considerations:
   - **(a)** Intended duration of the production bioreactor
   - **(b)** Target cell density of the production bioreactor
   - **(c)** Proclivity of the cell line and product toward filter fouling
   - **(d)** Stability of the product
2. Select an ATF exchange rate that meets the following criteria (typically, the vendor-suggested set point will come fairly close):
   - **(a)** ATF residence time should be below 75 s (this may need to be shorter for higher cell densities)
   - **(b)** ATF shear stress should be below some change-inducing threshold. For the cell line studied in the hydrodynamic stress section, this threshold would be 15,000 W/m^3, but this will likely depend on the shear sensitivity of the cell line under study
   - **(c)** Bioreactor residence time should be greater than or equal to the residence time at the current scale
3. If it is difficult to meet these different constraints, use other levers if necessary:
   - **(a)** Consider changing tubing diameters: larger diameters will decrease shear but increase dead volume and ATF residence time; smaller diameters will do the reverse
   - **(b)** Consider attaching two ATFs if one is not sufficient to meet the filter flux requirement
   - **(c)** Consider increasing the bioreactor dissolved oxygen set point if ATF residence time is a concern

While ATF filtration has many potential impacts on cell culture processes, the studies presented here have demonstrated how some of these effects can be quantified via both modeling and experiment, allowing for more robust, well-understood implementation and operation. ATF modeling and understanding will become even more valuable as bioreactor processes are intensified and pushed to higher cell densities, productivities, and volumes, leading to larger demands on filtration systems.

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NOMENCLATURES

A cross-sectional area

ATF	alternating tangential flow
CDF	cumulative distribution function
CHO	Chinese hamster ovary
CRBALO	cells reaching bulb at least once
CSPR	cell-specific perfusion rate
c_h	harvest concentration/activity
c_{O_2}	molar solubility of oxygen
c_r	bioreactor concentration/activity
D	diameter
DO	dissolved oxygen
E	filter exposure to protein and debris
EDR, ϵ	energy dissipation rate
f_{exch}	ATF bulb fraction exchanged each cycle
f_{O_2}	dissolved oxygen level
f	fanning friction factor
j	filter flux
L	length
LDH	Lactate dehydrogenase
μ	viscosity
N-1	seed train stage prior bioreactor
PES	polyethersulfone
ΔP_L	pressure drop across length L
Q	actual ATF exchange rate
Q_{input}	ATF exchange rate as inputed by user
q_{O_2}	cell-specific oxygen consumption rate
r	radial position
R	radius
Re	Reynolds number
ρ	density
SUB	single-use bioreactor
σ	sieving coefficient
t	time
t_{cycle}	ATF cycle time
$t_{deplete}$	time till oxygen depletion
t_{half}	ATF half-cycle time
TFF	tangential flow filtration
τ	ATF residence time
v	average velocity
V_{bulb}	volume of the ATF bulb
V_{filter}	volume of the ATF hollow-fiber filter
$V_{pump,0}$	assumed ATF pump volume
V_{pump}	actual ATF pump volume
V_{xfer}	volume of the ATF transfer line
x_v	viable cell density

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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